STRhub Verified
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Technical Reproducibility Validation Report

Hipstr
Version v0.7
Autosomal STR

Verification date	2026-06-24
Repository commit	b2033bfbb5cf55496b776463bdf2993fa763a4be
Attestation level	**Runs + Output Structure Validation**
Permalink	https://strhub.app/verified/hipstr-v0-7
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution — confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**5/5 gates passed — Runs + Output Structure Validation**
ATTESTATION LEVEL	**Runs + Output Structure Validation**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	Hipstr
Version	v0.7
Repository	https://github.com/tfwillems/HipSTR
Commit (pinned)	b2033bfbb5cf55496b776463bdf2993fa763a4be
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-06-24 19:56:25 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/28125530063

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the BAM, reference, and BED paths with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · Hipstr v0.7 · commit b2033bf
$ HipSTR \
--bams \
/data/in/input.bam \
--fasta \
/data/ref/hg38.fa \
--regions \
/data/in/regions.bed \
--str-vcf \
/data/out/result.vcf.gz \
--min-reads \
5 \
--use-unpaired \
--read-qual-trim \
! \
--def-stutter-model \
--max-str-len \
250
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Output Structure Validation	Output structure matches expected genotype-bearing format	PASS
Execution log	hipstr-v0-7.log-external.txt	

5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	result.vcf.gz
Format	VCF
Sequence records	23
Distinct STR loci	23
Total reads	8,230
Max depth (single locus)	523

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below.

Dataset	Illumina BAM (hg38) — NA12878 (autosomal, female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38

Loci tested	CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · SE33 · TH01 · TPOX · vWA

8. Verification Matrix

PASS	Reference dataset	Illumina BAM (hg38) — NA12878 (autosomal, female)
	README check (5/5)	Advisory — all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ **Runs end-to-end**

Hipstr v0.7 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

◆ **Produces valid output**

The tool generated a structurally valid VCF file with genotype calls across 23 target forensic STR loci, with a maximum read depth of 523 at D18S51.

◆ **No bundled demo data**

Hipstr does not include its own test or demo dataset. STRhub Verified supplied the GIAB NA12878 300× dataset (GRCh38) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix — Detected Markers with Read Depth

Hipstr v0.7 · verified 2026-06-24

LOCUS	READ DEPTH	LOCUS	READ DEPTH
D18S51	523	**D12S391**	361
FGA	475	**D1S1656**	360
vWA	465	**D8S1179**	349
PentaD	444	**D13S317**	321
D2S441	433	**D10S1248**	313
D3S1358	423	**D19S433**	300
D22S1045	416	**CSF1PO**	280
D7S820	412	**TH01**	278
PentaE	408	**D2S1338**	265
D5S818	400	**TPOX**	264
D16S539	374	**SE33**	0
D6S1043	366		